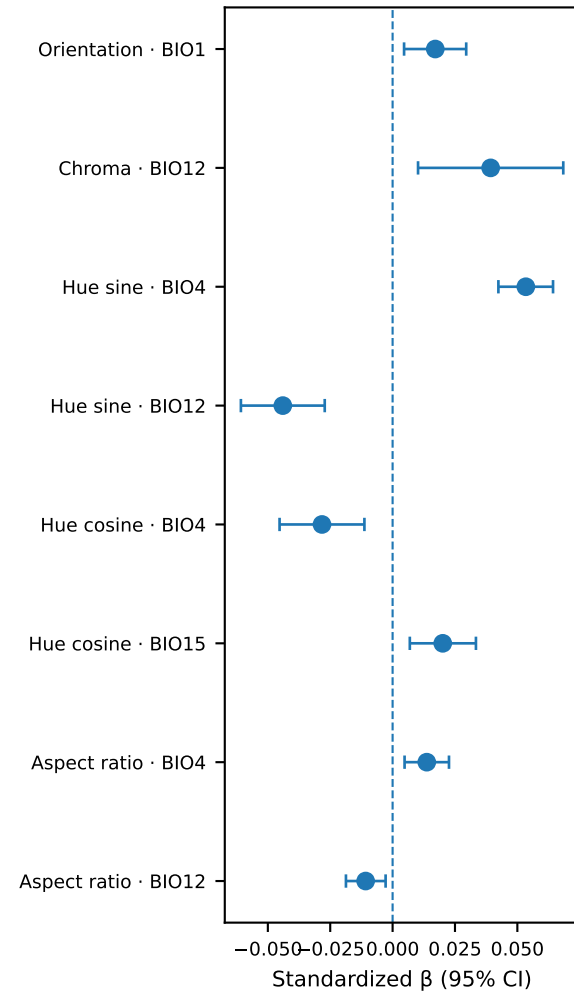


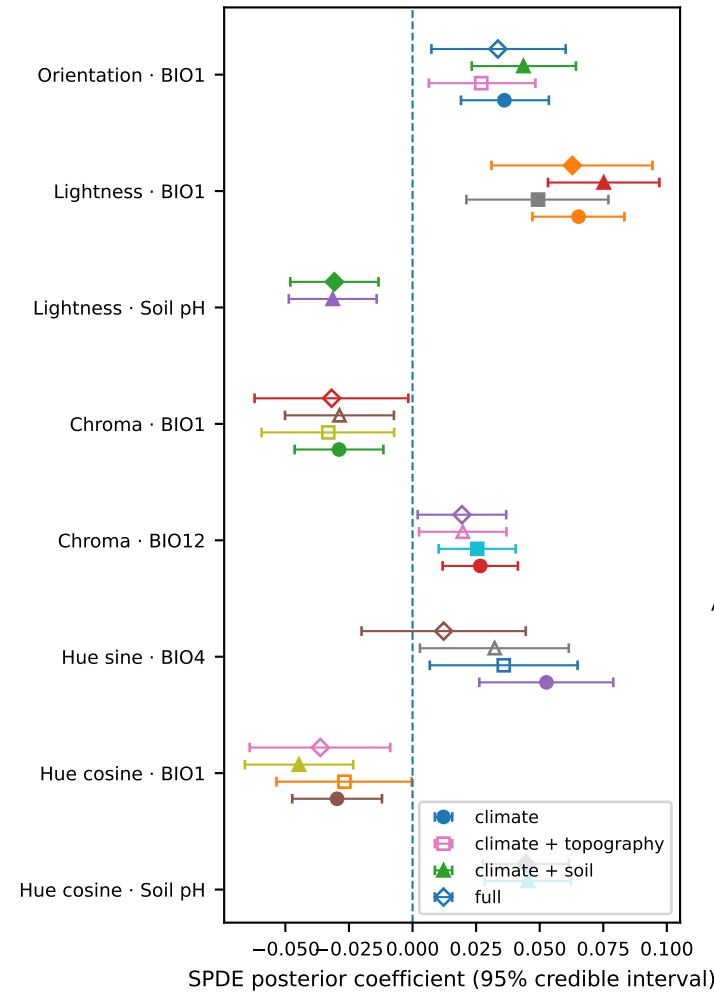
Effect sizes make the trait-specific environmental structure explicit

a Exhaustive within-taxon primary
8/36 BH-supported components

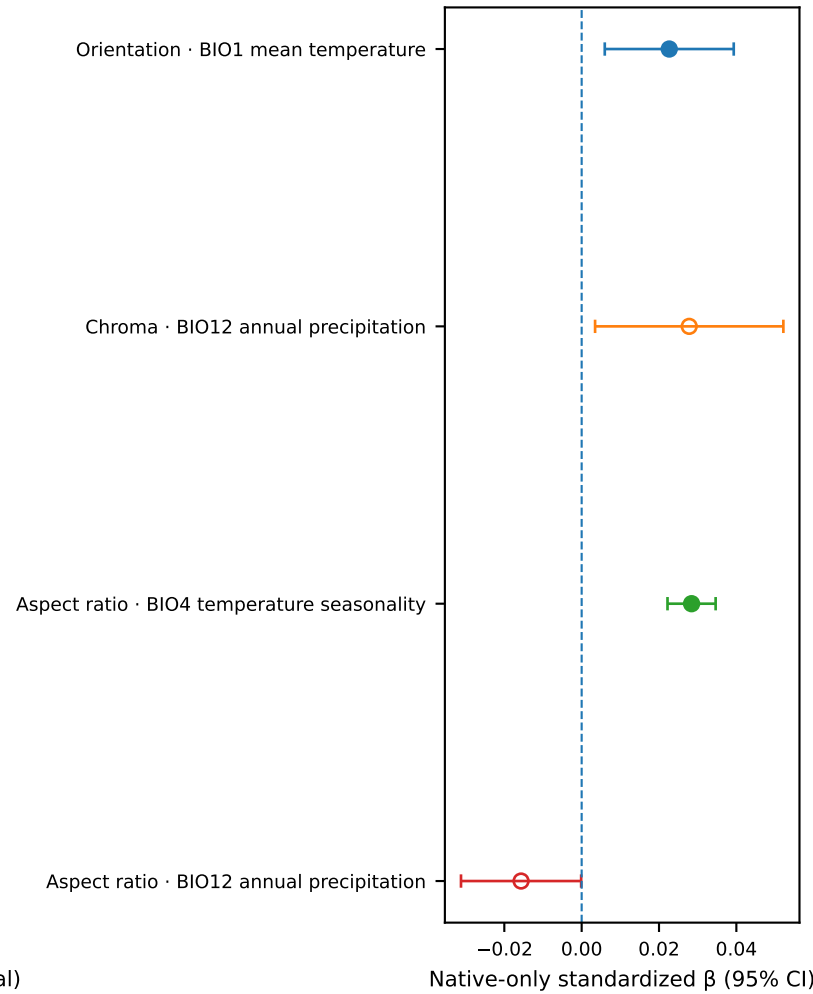


Hue sine/cosine are circular components and require joint interpretation.

b Grouped SPDE-INLA stable patterns
filled marker = global BH q < 0.05 in that model group



c Native-range restriction after timing/taxon audits
2/4 non-circular rows meet the locked rule



Filled: same sign and native-only BH q < 0.05. Open: same sign but BH failed. Stage and colour gates remain open.